

Materials science involve investigating the relationships that exist between the structure and properties of materials.

Materials engineering involve, on the basis of these structure property correlation, design/engineer the structure of a material to produce a predetermined set of properties.

From a functional perspective, the role of a materials scientist is to develop or synthesize new materials, whereas a materials engineer is called upon to create new products or systems using existing materials and to develop techniques for processing materials.

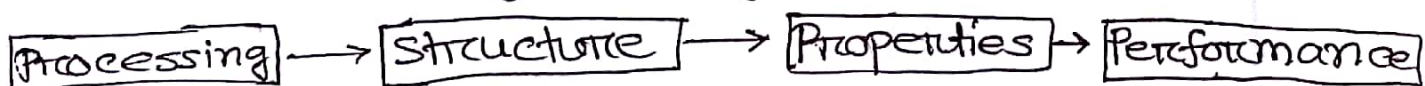
The structure of a material usually relates to the arrangement of its internal components. Subatomic structure involves electrons within

the individual atoms and interactions with their nuclei.

The notion of 'property' deserves elaboration. A property is a material trait in terms of the kind and magnitude of response to a specific imposed stimulus. Generally, definitions of properties are made ~~of~~ independent of material shape and size.

In addition to structure and properties, two other important components are involved in the science and engineering of materials - namely 'processing' and 'performance'. With regard to the relationship of these four components, the structure of a material will depend on how it is processed. Furthermore, a material's performance will be a function of its properties.

The four components of the discipline of materials science and engineering.



The relationship among these four components in terms of the design, production, and utilization of materials.

Ceramics

ceramics materials are inorganic materials which consist of metallic and nonmetallic elements chemically bonded together. Ceramics materials can be crystalline, non-crystalline or mixtures of both. Ceramics comes from Greece word "Keramikos" which means burnt stuff.

Classification of ceramics:

Classification based on composition:

- (1) Silicate Ceramics :- clay ceramics (with mullite - $3\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$), silica ceramics (with cordierite - $2\text{MgO} \cdot 2\text{Al}_2\text{O}_3 \cdot 2\text{SiO}_2$)
- (2) Oxide ceramics :-
 - (i) single oxide (Al_2O_3)
 - (ii) modified oxide (Zirconia toughened alumina)
 - (iii) mixed oxide (BaTiO_3).

(3) Non-oxide ceramics:- Carbon, SiC, BN.

(4) Glass ceramics:- partially crystallised glass. (SiO_2 - Li_2O , LAS, MAS.)

and another classification based on application:

(1) Traditional ceramics:- clay based

- products -
- (i) porcelain
 - (ii) sanitary ware
 - (iii) tiles
 - (iv) bricks
 - (v) refractories

(2) Natural ceramics:- rocks, and minerals, ice, bones.

(3) Cement:- a complex ceramics with many phases -

- (i) structural
- (ii) composite

(4)

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(4) Glasses: based on SiO_2 with ^{mechanical properties} additions to reduce ~~amp~~ or give special properties.

(i) containers

(ii) households

Glass $\rightarrow$ supercooled liquid.

(iii) optical glasses.

Any materials that has solidified and become rigid without forming a regular crystal structure is known as glass.

(5) High performance advanced

Ceramics: (i) cutting tool

(ii) sensor

(iii) grinding

(iv) laser

(v) bearing

(vi) superconductor

(vii) Insulator.

Properties of ceramics:

(1) High melting point and high refractoriness (except glass)

(2) Generally electrical and

thermal insulators

- ③ hard and strong with low plasticity
- ④ Low fracture toughness (brittle)
- ⑤ chemically inert.
- ⑥ Many are low cost (bricks)
- ⑦ wide range of appearance.
- ⑧ Opaque to light.
- ⑨ Resistance to high temperature

classification of polymers

- ① thermoplastic polymers
- ② thermosetting polymers
- ③ elastomers
- ④ biodegradable polymers
- ⑤ conductive polymers
- ⑥ smart polymers
- ⑦ liquid crystalline polymers
- ⑧ photoconductive polymers
- ⑨ piezoelectric polymers
- ⑩ shape memory polymers

Polymers

~~Generally~~ In other words

Polymers are substance made up of repeating structural units, each of which can be regarded as derived from a specific compound called a monomer.

And the word 'polymer' is the 'Greek' word: 'poly' means many and 'mer' means unit or parts. A polymer is a large molecule that comprises repeating structural units joined by covalent bonds.

Classification of polymers:-

- ① Classification based on source under:-
 - (i) Natural polymers:- proteins, cellulose, starch, rubber.

(ii) Semi-synthetic polymers: Silicom.

(iii) Synthetic polymers: Nylon, polyethene, PVC, Teflon.

✓ (2) Classification based on the structure of polymers:

(i) Linear ^{chain} polymers on structure: -
1. PVC, Polyethene.

(ii) Branched chain polymers: - polypropylene, glycogen.

(iii) Cross linked polymers: Bakelite, melamine, formaldehyde resin
(iv) Hyper branched polymer

✓ (3) Classification based on mode of polymerisation:

(i) Addition polymers: polyethene.

(ii) Condensation polymers: nylon 6,6.

4) Classification based on molecular forces:-

(i) Elastomers: buna-S, buna-N, neoprene.

(ii) Fibres: Polyamides, polyesters.

(iii) Thermoplastic polymers: polythene, polyester, PVC.

(iv) Thermosetting polymers: bakelite, urea-formaldehyde resins.

✓ 5) Classification based on the basis of types of monomers:-

(i) Homopolymer: polystyrene

(ii) Heteropolymer: butadiene-styrene rubber.

Properties of polymers:-

(i) Low density

(ii) Low efficient of friction

(iii) Good corrosion resistance

(iv) Good mould ability

(v) Excellent surface finish can be obtained.

(vi) Can be produced with close dimensional tolerances

~~(vii) Economical~~

(viii) Poor tensile strength

(ix) Low mechanical properties

(x) Poor temperature resistance

(xi) Can be produced transparent or in different colours.

Composites

Composite material is a material composed of two or more distinct phases (matrix phase and dispersed phase) and having bulk properties significantly different from those of any of the constituents.

Matrix is usually more ductile and less hard phase. It holds the dispersed phase and shares a load with it. And Dispersed phase is usually stronger than the matrix, therefore it is sometimes called reinforcing phase.

Many of common materials (metal alloys, doped ceramics, and

polymers mixed with additives) also have a small amount of dispersed phases in their structures, however they are not considered as composite materials since their properties are similar to those of their base constituents.

Classification of Composites:

(Ans)

There are two classification systems of composite materials. One of them is based on the matrix material and the second is based on the reinforcements.

(1) Classification of composites based on matrix material

① Metal Matrix Composites:

Metal matrix composites are composed of a metallic matrix (Al, Mg, Fe) and a dispersed ceramic (oxides, carbides) or metallic phase.

(ii) Ceramic Matrix Composites :-
Ceramic matrix composites are composed of a ceramic matrix and embedded fibres of other ceramic material (dispersed phase)

(iii) Polymer Matrix Composites :-
Polymer matrix composites are composed of a matrix from thermoset or thermoplastic and embedded glass, carbon, steel (dispersed phase)

(2) classification of composites based on reinforcements.

(1) Particle reinforced composites

(1) Fibre reinforced composites :-

a) Aligned fibre

b) Randomly fibre

(ii) structural reinforced composites:

a) Laminate composites :- Piling

of layers or lamina or panels of unidirectional composite materials or materials. The layers are stacked and subsequently joined together.

b) Sandwich composites :- Two external strong layers attached to a layer of less dense material having low stiffness and low strength.

Properties of composites are;

(1) Low density

(2) ~~Higher~~ ^{Higher} tensile strength

(3) Specific tensile strength

(4) Higher fatigue endurance

(5) Higher toughness than ceramics and glasses.

(6) Versatility and tailoring by design

(7) Low reusability

(8) Low recyclability.

Advanced materials

Materials that are utilized in high-technology applications are sometimes termed advanced materials. The advanced materials are typically traditional materials whose properties have been enhanced and also newly developed, high-performance materials.

Advanced materials include -

(1) Semiconductor

(2) Superconductor

(3) Bio-materials

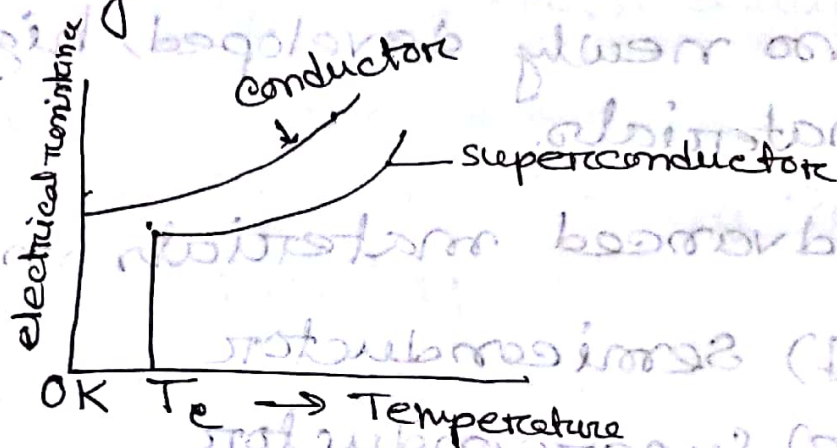
(4) Nano-materials

(1) Semiconductor:- 10-11 pages in details.

(2) Superconductor:- A superconductor is a material that can conduct electricity or transport electrons from one atom to another with no resistance. This means no heat, sound or any other form of

energy would be released from the material when it ~~is~~ has reached critical temperature (T_c) or the temperature at which the material becomes superconductive.

Unfortunately



unfortunately, most materials must be in an extremely low energy state in order to become superconductive. Research is underway to develop compounds that become superconductive at higher temperatures. Currently an excessive amount of energy must be used in the cooling process making superconductors inefficient and uneconomical.

mical, biotomoid, stromal gel, biotom

(3) Biomaterials:- A biomaterial is any substance that has been engineered to interact with biological systems for a medical purpose - either a therapeutic one (treat, augment, repair or replace a tissue function of the body) or a diagnostic one.

Biomaterials can be derived either from nature or synthesized in the laboratory using a variety of chemical approaches utilizing metallic components, polymers, ceramics or composite materials. They are used and adapted for a medical application, and thus comprises whole or part of a living structure or biomedical device which performs, augments or replaces a natural function. Such functions may be relatively passive, like being used for a heart valve, or may be bioactive with a more interactive functionality such as hydroxy-apatite

coated hip implants. Biomaterials are also used everyday in dental applications, surgery, and drug delivery.

(4) Nano-materials:- Nanomaterials describe, in principle, materials of which a single unit is sized between 1 and 1000 nanometres but is usually 1-100 nm.

Nanomaterials research takes a materials science based approach to nanotechnology, leveraging advances in materials metrology and synthesis which have been developed in support of micro and microfabrication research. Materials with structure at the nanoscale often have unique optical, electronic or mechanical properties.

Dimensions of nanomaterials:-

Materials that have one dimension in the nanoscale:- graphene, thin films

or surface coatings.

(2) two ~~dimension~~ dimension in nanoscale :- nanowires, and nanotubes.

(3) three dimension in nanoscale: colloids, quantum dots.

Nanomaterials are slowly becoming commercialized and beginning to emerge as commodities.

Factors affecting the selection of materials for engineering purposes:

(A) Properties of materials:

(1) Mechanical property

(2) Thermal property

(3) Electrical property

(4) Chemical property

(5) Radiation.

(B) Material performance requirements

(1) Functional requirements

(2) Processability requirements

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Materials selection

Materials selection is one of the foremost functions of effective engineering design as it determines the reliability of the design in terms of industrial and economical aspects. A great design may fail to be a profitable product if unable to find the most appropriate materials combination. So it is vital to know what the best materials for a particular design are.

Factors affecting the selection of materials for engineering purposes:

① Properties of materials:

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- (2) Thermal property
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- (4) Chemical property
- (5) Radiation.

② Material performance requirements:

- (1) Functional requirements
- (2) Processability requirements

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(3) Cost

(4) Reliability requirements

(5) Resistance to service conditions

(6) Safety

(7) Electrical condition

(8) Availability

(9) Disposability and

(10) Recyclability.

Applications of materials science and engineering.

There is a wide range of applications of materials science and engineering.

Traditional materials science and engineering focused on metallurgy, where at the effects of heat treating and doping to extract metal from the ground and give it specific properties. And think about the chemistry and physical process to get gold, iron and copper out of dirt in the ground. And the end products like spring steel, stainless steel, aircraft

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grade aluminium.

In aerospace, they need materials that are strong, even at extreme temperatures, usually special ceramics or composites.

In electronic materials, there is research in everything from OLED TV screens to quantum computing. Even the manufacture of tiny computer chips use lithography, which is an application of material science.

There are bio-materials, where research is being done to improve the quality of dental implants and hip replacements among other materials that would be in contact with, or embedded in the human body.

Building materials like Drywall can now be lighter weight, tougher and better at damping sound, and even abs-

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orbing volatile organic compounds.

Extending the life of a nuclear reactor comes down to inspecting all the materials, and checking if and why materials are degrading. It requires ~~to~~ experts to ensure the tight calculations are made ranging from corrosion to pressure to heat and mass transfer.

Polymers and composites are used to make ^{light} weight materials used in race cars and sports equipment.

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